

Stakeholder 15

BMS/HVAC operators

80 practical questions for bounded, evidence-aware building operation

OPERATING PRINCIPLE

Use verified current evidence to support authorised BMS/HVAC decisions. Sensors, integrations, point mappings and analytics are proposed until deployment, mapping, commissioning, permission and current health are independently verified. Missing or conflicting evidence stays visible.

NON-AUTHORITY

The service is read-only decision support. It does not issue control commands, change setpoints or schedules, start or stop plant, clear overrides, acknowledge or suppress alarms, bypass interlocks, certify indoor-air quality or safe operation, or replace authorised operators, engineers or emergency procedures.

Eight operator decision areas

- 1 Zone conditions, schedules and operating modes
- 2 Setpoints, overrides and control intent
- 3 Temperature, humidity, air quality and comfort evidence
- 4 Plant, valves, dampers, fans and equipment status
- 5 Alarms, faults, drift and sensor-quality diagnostics
- 6 Control optimisation and bounded operational recommendations
- 7 Maintenance, commissioning and shift handover
- 8 Energy, peak demand, resilience and seasonal performance

Scope and operating boundary

A practical catalogue for asking what current building evidence can support while keeping operational and safety authority with authorised roles.

PURPOSE

Operator-facing questions about schedules, control intent, environmental conditions, plant behaviour, alarms, maintenance, energy and resilience. Each record states the evidence, current owners, analysis operation, uncertainty, safe failure and the operator decision it can support.

EVIDENCE STATUS

A sensor, point, integration, topology link or analytic is not operational fact merely because it appears in a design or connector. Deployment, semantics, physical mapping, commissioning, permissions, time alignment and current health must be independently verified.

SPATIAL EVIDENCE

The May 2021 Abacws floor plan is historical spatial reference only. It cannot establish current room use, access, services, HVAC topology, hazards, closures or isolation boundaries. Current authorised records take precedence.

AUTHORITY BOUNDARY

Talking Abacws may explain evidence, compare live state with approved intent, expose uncertainty and suggest bounded verification. It must not write to the BMS, alter controls, release isolations, suppress alarms, certify IAQ/comfort/safe operation or replace competent professionals or emergency procedures.

Field guide

Each question card shows what is needed for a defensible answer before an operator acts on it.

1 BMS/HVAC operators question

A natural role-specific question tied to a real operating decision.

2 Sensors and telemetry

Proposed signals, modality/location, coverage, timing, aggregation, calibration or field-check status where applicable, clock health, uptime, gaps, drift and provenance.

3 Authoritative supporting data

Current owners and source precedence, with effective/observed/retrieved times, permissions, versions, staleness and conflicts.

4 Analysis required

Named operation type, decision interval, minimum evidence, method, uncertainty and explicit safe failure.

5 Answer boundary

What the service must not infer, disclose, promise, certify, authorise or control.

6 Why this is useful

The concrete operator investigation, escalation, handover, maintenance or engineering decision the evidence can support.

Analysis operation types

Lookup

retrieve a current authorised fact

Observation

state what verified evidence directly shows

Calculation

derive a deterministic quantity

Estimate

quantify with an explicit model or proxy

Forecast

project a future value with uncertainty

Comparison

contrast live/current evidence with an approved or trusted reference

Diagnosis

rank plausible explanations from evidence without claiming certainty

Recommendation

propose a bounded next assessment or action for an authorised person

Evidence hierarchy and time semantics

Operational answers should show who owns the evidence, when it applies, when it was observed and when it was retrieved.

Default precedence

- 1 Emergency, life-safety, permit-to-work and resilience instructions**
Always outrank routine optimisation or analytics.
- 2 Current approved source-of-record and control configuration**
Approved BMS schedules/configuration, CMMS, asset/service, energy and safety records.
- 3 Commissioned live telemetry**
Only when point semantics, mapping, timestamps, permissions and current health are verified.
- 4 Authorised operator and maintenance notes**
Useful context with clear author/time; they do not silently overwrite the source-of-record.
- 5 Historical references**
Older drawings and the May 2021 floor plan provide context only, never current operational authority.

Three times to preserve

Effective time

When a schedule, setpoint, configuration, work order, permission or operating rule is intended to apply.

Observed time

When the controller, sensor, meter or event occurred at source.

Retrieved time

When Talking Abacws obtained the record, exposing cache or connector delay and staleness.

Evidence, permissions and safe failure

Every answer must be permitted, technically interpretable and strong enough for the stated operator decision.

1 Permission

Confirm role permission for each BMS point, schedule, alarm, CMMS, energy, weather or change-control source before joining data. Technical reachability is not permission.

2 Mapping

Verify point/object identity, physical or logical location, served asset/zone relationship, engineering units, scaling and current mapping/configuration version.

3 Time

Check source clock, observed time, retrieved time, event ordering and clock alignment. Do not compare transitions or causal sequences across unsynchronised sources.

4 Sensor quality

For physical sensors/meters, use applicable calibration or field-check status, range/stuck/drift diagnostics and stated accuracy. Digital states require verified semantics and polarity, not calibration.

5 Coverage

Show expected versus valid samples, uptime, last-good time, gaps and aggregation window. Match sampling to the phenomenon: events for alarms, seconds/minutes for plant response, longer windows for drift and energy.

6 Safe failure

If required evidence is stale, conflicting, under-sampled, unhealthy, unmapped or permission-limited, answer unknown / insufficient evidence and name the missing proof. Do not fill gaps with an unlabelled model.

Permanent action boundary

The service does not issue BMS writes, change setpoints/schedules/priorities, start or stop plant, clear/extend overrides, acknowledge/suppress/reset alarms, bypass interlocks, authorise work/access/isolations, or certify IAQ, comfort, compliance or safe operation. Current authorised owners, competent professionals and emergency procedures remain authoritative.

BMS/HVAC OPERATORS QUESTION

BM-001

Which HVAC zones are not in the operating mode that their current approved schedule calls for right now?**Sensors and telemetry**

Signals / location: zone mode/status, controller occupancy-state, plant enable and schedule-reference points. Coverage / timing: every commissioned HVAC zone and its controlling schedule object; no inference from room occupants. 1-5 min observations; aggregation: 5 min status with 15 min persistence flag. Quality / provenance: controller clocks and mode enumerations must be verified; stale or unmapped zones remain unknown. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - schedule-to-live-mode reconciliation. Decision interval: every 5 min and on schedule transitions. Minimum evidence: current approved schedule version, verified zone-to-controller mapping and at least 3 healthy mode observations in 15 min. Method: derive expected mode from effective schedule/exception then compare with controller state and plant enable proof. Uncertainty: show mismatch age, number of supporting points and any schedule/controller conflict. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets the operator identify genuine schedule/mode mismatches before investigating plant or controls.

Answer boundary

Do not infer human presence from an occupied mode, or call a mismatch a fault without checking approved exceptions and interlocks. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-002

Which zones or systems are still running outside today's approved HVAC operating window?**Sensors and telemetry**

Signals / location: schedule objects, optimum-start/stop states, plant enable commands, run proof and terminal enable states. Coverage / timing: mapped AHUs, pumps and terminal groups with a verified relationship to an approved schedule. 1-5 min; aggregation: runtime intervals by system and current shift. Quality / provenance: exclude maintenance/test windows, failed run-proof points and controllers with unsynchronised clocks. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - approved-window overlap and exception detection. Decision interval: current shift, refreshed every 5 min. Minimum evidence: valid schedule/exception calendar plus 15 min of healthy command and run-proof evidence. Method: compare actual run intervals with approved windows and classify only unexplained overlap as candidate excess runtime. Uncertainty: report unmatched runtime duration and whether evidence is command-only, proof-only or both. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Supports rapid checks for unintended after-hours operation without disrupting legitimate exceptions.

Answer boundary

Do not assume that after-hours runtime is wasteful when an authorised extension, maintenance task, frost protection or emergency instruction may apply. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-003

Which systems started late or stopped late against today's approved schedule, and by how much?**Sensors and telemetry**

Signals / location: schedule transition timestamps, controller mode changes, plant enable and run-proof transitions. Coverage / timing: commissioned systems with reliable event timestamps and explicit scheduled start/stop semantics. event-level plus 1 min state samples; aggregation: per scheduled transition. Quality / provenance: clock synchronisation and event-ordering must be healthy; ignore transitions during controller restart or test mode. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: calculation - transition-latency calculation. Decision interval: after each scheduled start/stop and at shift assessment. Minimum evidence: one scheduled transition, corresponding controller/plant transition and trustworthy clocks. Method: calculate signed delay from approved effective transition to verified actual state, separating command delay from run-proof delay. Uncertainty: show timestamp resolution and whether the true transition may lie between samples. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps the operator distinguish schedule timing issues from downstream plant-response issues.

Answer boundary

Do not treat a late transition as equipment failure without checking optimum-start logic, safeties, interlocks or authorised delay. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-004

Which zones have not stabilised after the approved warm-up or cool-down period should have completed?**Sensors and telemetry**

Signals / location: zone temperature, active setpoint/band, operating mode, heating/cooling demand, terminal airflow and relevant plant enable proof. Coverage / timing: zones with commissioned temperature and demand points plus a verified service path. 1-5 min; aggregation: 15 min rolling stability and recovery trend. Quality / provenance: exclude sensors in known fault, recent maintenance or manual test; use only verified engineering units. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - post-transition settling assessment. Decision interval: 15 min after expected recovery completion, then every 10 min. Minimum evidence: current mode/setpoint, at least 30 min of healthy zone temperature and evidence that the serving system was enabled. Method: test whether temperature is within the approved band and rate-of-change has settled, comparing with trusted recovery history only when comparable. Uncertainty: report weather/load comparability and do not treat historical recovery time as a guarantee. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Shows where an operator may need to investigate slow recovery, load changes or service constraints.

Answer boundary

Do not declare discomfort, unsafe conditions or inadequate design solely from BMS temperature response. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-005

Where does the approved schedule call for occupied mode while the local controller or terminal remains inhibited, setback or unoccupied?**Sensors and telemetry**

Signals / location: schedule state, zone occupancy-mode input used by the controller, terminal enable/inhibit, local controller mode and actuator-demand points. Coverage / timing: mapped zone controllers and terminal units with known control priority and inheritance. 1-5 min; aggregation: current state with 15 min persistence. Quality / provenance: verify meaning of inhibit/setback enumerations and source priority; do not infer real occupancy. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - control-state inheritance check. Decision interval: every 5 min during scheduled occupied periods. Minimum evidence: approved occupied schedule plus healthy local-mode/enable telemetry and control-priority metadata. Method: trace schedule -> supervisory command -> local mode/inhibit and flag persistent state breaks. Uncertainty: show which step is missing and whether a local safety or authorised override could explain it. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps the operator locate where approved occupied intent is being lost in the control chain.

Answer boundary

Do not infer that a room is empty or occupied, or override a local safety/inhibit state. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-006

What after-hours HVAC extensions are active now, who or what authorised them, and when do they expire?**Sensors and telemetry**

Signals / location: after-hours request state, extension timer, affected schedule/zone objects, plant enable and expiry timestamp where exposed. Coverage / timing: all verified extension mechanisms visible to BMS, using role or request class rather than personal identity. event-level and 1-5 min state; aggregation: one record per active extension. Quality / provenance: authorisation source and expiry semantics must be known; omit personal requester details unless operationally required and permitted. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - active-extension inventory and expiry validation. Decision interval: on request and every 5 min while active. Minimum evidence: active extension state, verified affected scope, current effective authorisation and expiry/end condition. Method: join extension event to affected schedule objects and show remaining authorised window; flag missing/expired governance metadata. Uncertainty: show unknown when requester/expiry provenance is unavailable rather than guessing. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives the operator a clean list of legitimate temporary runtime and items needing authorised follow-up.

Answer boundary

Do not disclose unnecessary personal details, extend/cancel the request, or decide that an extension is no longer justified. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-007

Are any systems in holiday, weekend or closure mode when the current approved calendar says they should be operating normally?**Sensors and telemetry**

Signals / location: calendar exception state, day-type/holiday flag, schedule outputs, system mode and plant enable proof. Coverage / timing: systems whose schedules inherit from a verified calendar or day-type object. on calendar/schedule events plus 5 min state; aggregation: per system and active exception. Quality / provenance: calendar timezone, effective dates and inheritance must be verified; current closure/access records take precedence over historical room use. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - calendar-exception consistency check. Decision interval: at day start, after calendar edits and every 15 min during mismatch. Minimum evidence: current approved exception calendar, schedule inheritance and live mode state. Method: compare current date/day-type and approved exception with resulting system mode, tracing inherited schedules when possible. Uncertainty: state whether mismatch may arise from approved closure, local exception or stale calendar data. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps avoid whole-area HVAC being left in the wrong day type after calendar changes.

Answer boundary

Do not assume a building area should be open because of the May 2021 floor plan or routine timetable. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-008

Which systems are in frost protection, purge, smoke-related, night setback or another exceptional mode, and what evidence explains why?**Sensors and telemetry**

Signals / location: system mode, safety/interlock states exposed for read-only use, relevant temperature/air-quality triggers, timer states and plant proof. Coverage / timing: only systems with commissioned exceptional-mode semantics and permissions appropriate to the operator role. event-level and 1 min while exceptional; aggregation: current event episode. Quality / provenance: life-safety interfaces may be incomplete or restricted; absence of a BMS point is not evidence that a safety condition is absent. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - exceptional-mode episode reconstruction. Decision interval: immediate read-only status and at shift handover. Minimum evidence: verified mode code, start time and at least one authorised trigger/interlock source or explicit statement that cause is unavailable. Method: trace mode entry against available trigger, timer and interlock events without bypassing source-of-record safety systems. Uncertainty: report missing cause explicitly and treat safety-system evidence as authoritative over analytics. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets the operator understand why routine HVAC behaviour may be intentionally suspended and escalate appropriately.

Answer boundary

Do not interpret or clear a life-safety state, promise normal operation, or expose restricted safety details beyond permission. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-009

Which zones or systems have changed operating mode unusually often during this shift?**Sensors and telemetry**

Signals / location: time-stamped mode transitions, controller restart/status, schedule events, override events and relevant demand states. Coverage / timing: commissioned controllers with stable mode coding and event history. event-level; aggregation: transition counts per hour and episode duration. Quality / provenance: deduplicate retransmitted events and separate controller restarts from genuine mode changes. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: calculation - mode-transition frequency and sequence analysis. Decision interval: rolling 8-12 h shift window, refreshed every 15 min. Minimum evidence: at least 4 h of valid event history or 10 transitions, plus schedule/override context. Method: compare transition rate with a trusted same-system baseline and identify sequences tied to schedules, overrides or restarts. Uncertainty: show baseline sample size; do not label statistical rarity as a control fault. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Directs attention to unstable operating patterns that may deserve a deeper control or sensor check.

Answer boundary

Do not diagnose hunting or equipment failure solely from mode-change count without control-loop evidence. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-010

What is the current HVAC operating-mode picture across verified AHUs, terminal groups and their served zones?**Sensors and telemetry**

Signals / location: AHU mode/run proof, terminal enable/mode, zone mode and verified service-association identifiers. Coverage / timing: only current commissioned service relationships; historical floor-plan adjacency is not used to invent HVAC topology. 1-5 min; aggregation: current status snapshot with age of each point. Quality / provenance: service links and point mappings must have explicit provenance; partial paths stay partial. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS schedule/control-mode owner, the approved opening-hours or exception-calendar owner, and the current HVAC zone/service-mapping owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - hierarchical operating-state aggregation. Decision interval: on request and every 5 min for an operator dashboard. Minimum evidence: healthy top-level plant and terminal/zone state for each displayed branch plus current service mapping. Method: group by verified service path and show mixed/unknown states instead of forcing one building-wide label. Uncertainty: display coverage percentage and age of the oldest included observation. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides a reliable shift-level overview without hiding gaps in topology or telemetry.

Answer boundary

Do not infer current room use, access or service relationships from the May 2021 floor plan. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-011

Which zone, terminal or plant controls currently have an active manual, timed or software override?**Sensors and telemetry**

Signals / location: override flag/source/priority, commanded value, normal value, start time, expiry and affected control point. Coverage / timing: all commissioned writable points where read-only override metadata is available; no credentials or personal operator data. event-level plus 1-5 min state; aggregation: one active-override record per point. Quality / provenance: priority arrays and vendor-specific override semantics must be mapped correctly; missing expiry is not assumed permanent. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - active-override inventory. Decision interval: on request, every 5 min, and at shift handover. Minimum evidence: verified override state and control priority plus current approved normal value or strategy reference. Method: identify override scope, age, source class and expiry; distinguish local hardware, BMS software and schedule-level overrides. Uncertainty: show unknown source/expiry explicitly and do not infer operator intent from the overridden value. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets the operator see temporary control states that can mask normal sequence behaviour.

Answer boundary

Do not remove an override, reveal unnecessary operator identity, or decide whether the override remains authorised. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-012

Which active overrides have expired, are close to expiry, or have no verified end condition?**Sensors and telemetry**

Signals / location: override start/expiry timer, current override flag, priority/source and affected point value. Coverage / timing: active overrides with verified timer semantics and current control mapping. 1-5 min plus expiry events; aggregation: remaining duration buckets. Quality / provenance: controller time must be synchronised and "no expiry" must be distinguished from missing metadata. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - expiry-state classification. Decision interval: every 5 min and 30 min before verified expiry. Minimum evidence: active override, trustworthy current time and verified expiry/end-condition semantics. Method: calculate remaining authorised duration, flag passed expiry still active, and separate deliberate indefinite overrides from unknown expiry. Uncertainty: report timer resolution and uncertainty if time sources disagree. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Supports timely authorised assessment before temporary control states become forgotten.

Answer boundary

Do not automatically clear, extend or renew an override, or claim an expired override is unauthorised without checking the source-of-record. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-013

Do current occupied and unoccupied zone setpoints match the approved control strategy for this season and mode?**Sensors and telemetry**

Signals / location: effective heating/cooling setpoints, mode, season/changeover state, local trim and supervisory limits. Coverage / timing: zones with commissioned setpoint semantics and a current approved strategy reference. 1-5 min; aggregation: current value plus change history for the shift. Quality / provenance: distinguish effective setpoint from configured default, local trim and calculated reset value. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - setpoint-to-strategy conformance check. Decision interval: every 15 min and after approved configuration changes. Minimum evidence: approved strategy/version, effective mode and healthy current setpoint points. Method: compare effective setpoints with mode- and season-specific permitted bands, tracing reset/local trim where documented. Uncertainty: show strategy version and do not assume a generic comfort band when none is approved. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps detect configuration drift while respecting intentional reset and local adjustment logic.

Answer boundary

Do not declare a setpoint appropriate for occupants or alter it; comfort and safety judgements remain with authorised roles. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-014

Where are heating and cooling targets, deadbands or staging thresholds overlapping or closer than the approved control intent?**Sensors and telemetry**

Signals / location: effective heating/cooling setpoints, deadband parameters, staging thresholds, mode and local trim/reset outputs. Coverage / timing: zones and plant loops with explicit documented control bands. 1-5 min; aggregation: current band geometry plus hourly persistence. Quality / provenance: use the correct control units and distinguish zone targets from supply-water/air reset thresholds. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - deadband/threshold geometry validation. Decision interval: every 15 min and after configuration change. Minimum evidence: current approved band rules plus verified effective heating/cooling targets or thresholds. Method: calculate separation and overlap under the active mode, flag only values outside the documented allowable relationship. Uncertainty: include point precision and configurable tolerance around equality. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives the operator a precise configuration check before investigating simultaneous heating and cooling.

Answer boundary

Do not conclude that simultaneous energy use is occurring without actuator/plant evidence, or change the bands. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-015

Which overrides or non-default control priorities have persisted across a shift change?**Sensors and telemetry**

Signals / location: override state, priority/source, start time, expiry, affected point, operator shift boundary and related work-order reference where present. Coverage / timing: commissioned override-capable controls plus authorised shift/maintenance records. event-level; aggregation: active items crossing each shift boundary. Quality / provenance: shift timestamps and controller clocks must align; do not expose personal details beyond operational need. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: calculation - cross-shift persistence check. Decision interval: at each handover and on demand. Minimum evidence: active override/non-default priority before and after the shift boundary with provenance. Method: identify items that remained active through handover and join to authorised maintenance/change records when available. Uncertainty: show whether absence of a work-order link is missing data rather than proof of poor practice. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Prevents long-lived temporary states from disappearing from operational awareness at handover.

Answer boundary

Do not assign blame, infer operator performance, or remove the persistent state. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-016

Are any local zone requests or wall-controller trims currently outside the centrally approved allowable range?**Sensors and telemetry**

Signals / location: local trim/request value, supervisory min/max limits, effective setpoint, mode and controller clamp status. Coverage / timing: zones with privacy-neutral control requests; no user identity or room-activity inference. 1-5 min; aggregation: current out-of-range or clamped episodes. Quality / provenance: map local request scale correctly and distinguish a requested value from the effective clamped setpoint. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: calculation - request-versus-limit comparison. Decision interval: every 5 min during occupied schedules. Minimum evidence: healthy local request, approved supervisory limits and effective setpoint/trim result. Method: compare request to limits and determine whether the controller clamps, rejects or passes it according to documented logic. Uncertainty: report sensor/controller precision and any unknown local scaling. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators separate unusual local requests from faults in supervisory limit enforcement.

Answer boundary

Do not infer who made the request, why they made it, or alter local/supervisory limits. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-017

Where are heating and cooling control outputs simultaneously active for long enough to merit investigation?**Sensors and telemetry**

Signals / location: zone heating/cooling demand, valve/damper outputs, reheat command, cooling command, supply condition and mode. Coverage / timing: zones or AHU sections with independently mapped heating and cooling outputs. 1 min; aggregation: 5-15 min overlap episodes and daily duration. Quality / provenance: exclude changeover transients, defrost, dehumidification/reheat and documented sequence exceptions. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - simultaneous-output episode detection. Decision interval: every 5 min with daily assessment. Minimum evidence: at least 5 min of healthy heating and cooling output evidence plus current mode/sequence context. Method: detect overlap above documented output thresholds and classify known sequence exceptions before flagging candidates. Uncertainty: show duration, output magnitude and any missing heat-transfer proof. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Prioritises genuine control conflicts for operator or controls-engineer assessment.

Answer boundary

Do not equate command overlap with wasted energy or a fault without checking the approved sequence and physical response. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-018

After an approved schedule or setpoint change, which intended controllers have not adopted the new effective value?**Sensors and telemetry**

Signals / location: source configuration version, distribution/update events, controller effective value, last-write/readback timestamp and communication health. Coverage / timing: controllers explicitly in the approved change scope. event-level plus 1-5 min readback; aggregation: propagation status per target controller. Quality / provenance: version identifiers and controller addressing must be verified; retries must not be mistaken for multiple intended changes. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: observation - configuration-propagation verification. Decision interval: within the approved deployment window and for 30 min afterward. Minimum evidence: approved change record, target list, source version and verified readback from each target. Method: compare intended version/value to effective readback and classify received, pending, rejected, unreachable or unknown. Uncertainty: show communication delay and do not assume a missing readback means the controller retained the old value. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Shows exactly where an approved change may not have propagated so the operator can escalate safely.

Answer boundary

Do not re-send configuration, force a write, or certify successful commissioning from BMS readback alone. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-019

Which control loops are spending sustained time at output limits without achieving their current target?**Sensors and telemetry**

Signals / location: loop setpoint, process value, controller output, actuator feedback where available, mode and saturation/limit flags. Coverage / timing: commissioned PID or staged loops with known output scaling and target semantics. 10-60 s for fast plant loops; 1-5 min for slow zone loops; aggregation: 15 min saturation episodes. Quality / provenance: sampling must match loop dynamics; exclude disabled/manual/test states and verify output limits. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - loop saturation and unmet-target detection. Decision interval: rolling 15-30 min. Minimum evidence: healthy setpoint, process value and output for at least three expected loop time constants or 15 min, whichever is longer. Method: measure time at limits and persistent error, then correlate with actuator feedback/plant availability when available. Uncertainty: report lack of actuator/plant proof and avoid causal diagnosis from loop data alone. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps distinguish a control loop asking for maximum effort from one actually delivering the expected response.

Answer boundary

Do not retune the loop, force an actuator, or conclude the controlled equipment is undersized. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-020

For this zone, can I trace current control intent from schedule and mode through effective setpoints to actuator demand and measured response?**Sensors and telemetry**

Signals / location: schedule state, mode, effective setpoints, reset/trim values, loop outputs, actuator command/feedback and zone process values. Coverage / timing: one selected zone and its verified supervisory-to-terminal control chain. event history plus 1-5 min state; aggregation: ordered trace for the selected time window. Quality / provenance: every link must have known semantics and current mapping; missing links remain explicit gaps. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls/configuration owner, change-control owner, and current commissioning or local-controller configuration owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - control-intent provenance trace. Decision interval: on demand for a current issue or specified historical window. Minimum evidence: current approved schedule/strategy plus healthy observations at each available control-chain step. Method: construct an ordered causal trace of intended state and observed response without inventing missing topology or logic. Uncertainty: label inferred sequence relationships separately from documented ones and show timestamp alignment limits. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides an operator-friendly evidence chain for troubleshooting a zone without hiding missing control context.

Answer boundary

Do not claim undocumented logic, infer occupant behaviour, or authorise a change based on the trace. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-021

Which zones have remained outside their current approved temperature band long enough to require operator attention?**Sensors and telemetry**

Signals / location: zone temperature, effective heating/cooling band, operating mode and relevant supply/terminal demand points. Coverage / timing: commissioned zone-temperature locations with current service mapping. 1-5 min; aggregation: 15 min persistence and 1 h context. Quality / provenance: exclude sensors with calibration/fault flags or implausible jumps; sensor location may not represent the whole room. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - band-excursion duration analysis. Decision interval: every 5 min during applicable operating modes. Minimum evidence: healthy temperature series, approved band for the active mode and at least 15 min of observations. Method: calculate excursion magnitude/duration and compare with mode transitions before ranking. Uncertainty: show sensor accuracy/location limitations and whether the band itself is currently authorised. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets operators distinguish sustained thermal issues from brief, expected transients.

Answer boundary

Do not certify comfort, habitability or safe temperature, or infer individual occupant experience from a zone sensor. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-022

Which zones are recovering from setback more slowly than their own recent verified warm-up or cool-down pattern?**Sensors and telemetry**

Signals / location: zone temperature, effective target, mode transition, heating/cooling demand, supply condition and outdoor temperature where verified. Coverage / timing: zones with enough comparable historical recovery episodes and stable sensor mapping. 1-5 min; aggregation: one recovery episode with 5 min trend. Quality / provenance: compare only episodes with broadly similar mode, season and outdoor conditions; exclude maintenance/test periods. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - recovery-rate comparison. Decision interval: during each warm-up/cool-down episode. Minimum evidence: current recovery plus at least 5 trusted comparable past episodes or a documented commissioned baseline. Method: estimate time-to-band and temperature slope, then compare with matched historical episodes without treating history as design intent. Uncertainty: show comparison count and weather/load mismatch; return insufficient evidence when comparability is weak. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps an operator decide whether slow recovery is unusual enough to investigate today.

Answer boundary

Do not automatically advance start time or claim insufficient plant capacity. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-023

Where is relative humidity outside the approved operating range, and is the excursion persistent or sensor-limited?**Sensors and telemetry**

Signals / location: zone RH, temperature, mode, supply-air RH/dew point where available and sensor health/calibration status. Coverage / timing: commissioned RH sensors in zones or air systems where an approved RH criterion exists. 1-5 min; aggregation: 15 min persistence and 24 h episode summary. Quality / provenance: RH sensors require calibration awareness and can be sensitive to local placement; no criterion means no compliance judgement. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - RH range and persistence analysis. Decision interval: every 10 min and on sustained excursion. Minimum evidence: healthy RH/temperature series, approved range and at least 15 min of valid data. Method: calculate excursion magnitude/duration, screen for obvious sensor anomalies and compare supply/zone signals when available. Uncertainty: show sensor accuracy, missing criterion and location representativeness. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Supports targeted checks for persistent humidity control issues without overclaiming what RH telemetry proves.

Answer boundary

Do not certify moisture safety, mould absence, material protection or occupant health. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-024

Which commissioned CO2-monitored zones show sustained elevated CO2 during an approved occupied period?**Sensors and telemetry**

Signals / location: zone CO2, schedule/mode, ventilation airflow or damper position where verified, and sensor calibration/health. Coverage / timing: only zones with commissioned CO2 sensors; any occupancy evidence must be anonymous zone-level state/count and is optional. 1-5 min; aggregation: 15 min median plus 1 h trend. Quality / provenance: CO2 calibration, sensor age and placement must be known; scheduled occupied mode is not proof of people being present. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: observation - threshold/trend screening against the current approved operational criterion. Decision interval: every 5 min during applicable schedules. Minimum evidence: healthy CO2 data for at least 15 min, current criterion and ventilation-mode context. Method: report sustained level/trend and available ventilation response; use occupancy-independent interpretation unless authorised anonymous counts exist. Uncertainty: show criterion source and sensor uncertainty; no healthy CO2 sensor -> not instrumented. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets operators identify zones where ventilation response or sensing deserves prompt investigation.

Answer boundary

Do not certify indoor-air quality, ventilation compliance, health risk or actual occupancy. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-025

Where is the environmental evidence too incomplete to judge a reported thermal or air-quality issue from BMS data alone?**Sensors and telemetry**

Signals / location: available temperature, RH, CO2 and relevant ventilation/supply points plus each point's health, age and calibration status. Coverage / timing: the selected complaint/issue zone and time window; no speech, identity or research-content data. native sampling; aggregation: coverage matrix by variable and time. Quality / provenance: missing variables, uncommissioned sensors and poor placement are treated as evidence gaps, not normal conditions. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - evidence-sufficiency assessment. Decision interval: on demand for a current or historical issue. Minimum evidence: current zone mapping and point health plus at least one valid environmental signal for the requested interval. Method: score coverage by variable, time completeness, calibration status and service context; identify what cannot be concluded. Uncertainty: state coverage percentage and explicit unknowns rather than a single confidence score if dimensions conflict. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Tells the operator whether BMS evidence is adequate for diagnosis or whether a physical check/other evidence is needed.

Answer boundary

Do not fill gaps by proxy from nearby rooms unless a documented representative relationship exists, or certify comfort/IAQ. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-026

Which zones show repeated temperature or humidity excursions at roughly the same time on multiple comparable days?**Sensors and telemetry**

Signals / location: zone temperature/RH, schedule/mode transitions, relevant supply condition and outdoor weather where verified. Coverage / timing: commissioned sensors with stable mapping over the comparison period. 1-5 min; aggregation: 15 min excursion episodes aligned by time-of-day. Quality / provenance: exclude days with known maintenance, closures or configuration changes; verify clocks and daylight-saving handling. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - recurring diurnal-pattern detection. Decision interval: daily assessment over the previous 2-6 weeks. Minimum evidence: at least 5 comparable days with 80% valid coverage in the relevant window. Method: detect repeated excursion windows and compare against schedule/plant transitions and weather-matched days. Uncertainty: show number of comparable days and false-pattern risk when the sample is small. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators find schedule, solar/load or control interactions that are easy to miss in single-day views.

Answer boundary

Do not infer a behavioural cause, room use or occupant activity from recurring environmental patterns. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-027

Where do nearby or paired environmental sensors disagree enough to justify checking placement, stratification or sensor health?**Sensors and telemetry**

Signals / location: paired temperature/RH/CO2 sensors with known locations, calibration status and timestamps. Coverage / timing: only sensor pairs explicitly documented as comparable; physical proximity alone is not sufficient. 1-5 min; aggregation: 15 min paired differences and 24 h distribution. Quality / provenance: account for declared sensor accuracy and different heights/locations; do not use the historical floor plan to invent comparability. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: calculation - paired-sensor difference analysis. Decision interval: every 15 min with daily assessment. Minimum evidence: two healthy comparable sensors with at least 30 paired samples and known accuracy/location metadata. Method: calculate bias, spread and persistence beyond combined tolerance; separate constant offset from load-dependent difference. Uncertainty: report pair comparability and combined measurement uncertainty. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Directs maintenance effort toward plausible sensor or placement issues instead of treating every disagreement as a zone fault.

Answer boundary

Do not decide which sensor is correct without a trusted reference or physical check. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-028

Which zones currently have low-confidence environmental evidence because key sensors are stale, unhealthy, sparse or poorly mapped?**Sensors and telemetry**

Signals / location: point health, last-seen age, calibration status, sample completeness and current zone/service mapping for environmental sensors. Coverage / timing: all environmental points the viewer is permitted to see, health events plus native samples; aggregation: zone-level evidence-quality summary every 15 min. Quality / provenance: a communicating point can still be semantically wrong; mapping and commissioning status are separate quality dimensions. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - environmental evidence-quality scoring by dimension. Decision interval: every 15 min and before using data for diagnosis. Minimum evidence: point inventory, current mapping, last-seen, calibration/maintenance status and recent completeness. Method: classify freshness, completeness, mapping and calibration separately and identify the limiting dimension. Uncertainty: avoid one opaque confidence number; show each evidence limitation. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Warns the operator when the dashboard looks complete but the underlying evidence is not trustworthy enough for a conclusion.

Answer boundary

Do not treat missing sensors as normal conditions or use low-confidence evidence to certify comfort/IAQ. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-029

For a reported comfort issue in a known zone and time window, what BMS evidence can we reconstruct without identifying anyone?**Sensors and telemetry**

Signals / location: zone temperature/RH/CO2 where commissioned, schedule/mode, setpoints, ventilation/terminal demand and relevant alarms. Coverage / timing: the authorised zone and time interval only; no personal identity, speech, device, network-content or productivity data. native event/sample timing; aggregation: 5 min aligned timeline. Quality / provenance: do not widen the query to adjacent spaces unless service mapping justifies it; preserve data gaps. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - privacy-preserving event reconstruction. Decision interval: on demand for the specified issue window. Minimum evidence: verified zone/time, healthy points covering at least 70% of the interval, and current mapping/strategy context. Method: align environmental, control and alarm events into a chronology and distinguish observations from hypotheses. Uncertainty: show coverage and sensor representativeness; complaint details remain outside the BMS evidence unless separately authorised. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives the operator an evidence-based timeline for troubleshooting while keeping personal information out of the analysis.

Answer boundary

Do not identify a complainant, infer behaviour or productivity, or certify whether the complaint is valid. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-030

Do any monitored zones show a temperature-humidity combination that warrants a condensation-risk check by a competent person?**Sensors and telemetry**

Signals / location: zone temperature/RH, supply-air dew point where available, nearby surface temperature only where a commissioned sensor exists, and outdoor condition. Coverage / timing: zones with healthy temperature/RH; surface-risk screening only where actual surface-temperature evidence exists. 1-5 min; aggregation: 15 min dew-point trend. Quality / provenance: RH accuracy strongly affects dew-point calculation; wall/window surface temperatures cannot be guessed from air temperature. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS environmental-point owner, the owner of approved environmental criteria, calibration/maintenance records, and current zone/service mapping. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - dew-point and margin screening. Decision interval: every 15 min during sustained high RH or low surface margin. Minimum evidence: healthy temperature/RH for at least 15 min and, for surface margin, a verified surface-temperature measurement. Method: calculate air dew point and observed surface-to-dew-point margin; otherwise report only an air-moisture screening indicator. Uncertainty: propagate sensor uncertainty into dew-point/margin and return insufficient evidence without surface data. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators decide where a physical inspection or building-fabric assessment is justified.

Answer boundary

Do not certify absence/presence of condensation, mould or material damage, or infer hidden surface temperatures. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-031

Which AHUs are running now, what mode is each in, and which run proofs support that status?**Sensors and telemetry**

Signals / location: AHU enable/command, fan run proof, VFD speed, mode, supply/return conditions and key interlock states exposed read-only. Coverage / timing: all mapped AHUs in the authorised BMS view. 10-60 s for fan/VFD state; 1-5 min for air conditions; aggregation: current status with 15 min context. Quality / provenance: command without proof is not running; mode codes and interlocks must be mapped to current O&M; semantics. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M;/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - AHU operating-state synthesis. Decision interval: every 1-5 min. Minimum evidence: healthy command and at least one independent run proof or explicit statement that proof is unavailable. Method: combine command, proof, speed and mode to classify running, commanded-not-proven, stopped, inhibited or unknown. Uncertainty: show proof type and data age; do not infer mechanical condition from command alone. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides a dependable plant overview that separates commanded state from verified operation.

Answer boundary

Do not start/stop an AHU, bypass an interlock or declare it safe to operate. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-032

Which supply or extract fans are commanded on but lack verified evidence that they are actually running?**Sensors and telemetry**

Signals / location: fan command, run/status contact, VFD running/frequency, motor current or differential pressure where commissioned. Coverage / timing: mapped fans with at least one documented run-proof mechanism. 5-30 s; aggregation: 2-5 min command/proof mismatch episode. Quality / provenance: verify proof polarity and VFD status semantics; a pressure signal may be system-level rather than fan-specific. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M;/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - command-versus-run-proof reconciliation. Decision interval: every 1 min while commanded on. Minimum evidence: continuous on-command plus healthy proof point(s) for at least the documented start delay plus 60 s. Method: apply start-delay allowance then compare command with independent proof; rank stronger when multiple proof types agree. Uncertainty: state whether evidence is electrical, VFD, status contact or pressure-based. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators identify fans needing prompt inspection without confusing missing proof with confirmed failure.

Answer boundary

Do not declare motor failure or authorise restart/isolation from BMS evidence alone. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-033

Which dampers have a persistent mismatch between commanded position and verified feedback?**Sensors and telemetry**

Signals / location: damper command, position feedback, end-switches where present, mode and relevant airflow/pressure. Coverage / timing: mapped modulating or two-position dampers with commissioned feedback semantics. 10-60 s; aggregation: 5 min mismatch episodes. Quality / provenance: account for actuator travel time, deadband and feedback resolution; do not assume 0/100% means physically sealed/open. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - command-feedback tracking analysis. Decision interval: every 1-5 min while command changes or demand persists. Minimum evidence: healthy command and feedback across at least one movement or 5 min steady request. Method: calculate position error after travel allowance and test whether airflow/pressure provides corroborating physical response. Uncertainty: show tolerance and whether feedback is direct position or inferred end-switch state. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives the operator evidence to separate control demand, feedback problems and possible mechanical sticking.

Answer boundary

Do not force damper movement, diagnose linkage failure conclusively or override smoke/fire-related damper logic. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-034

Which heating or cooling valves have stayed near an end stop for a long period, and is the served condition still unmet?**Sensors and telemetry**

Signals / location: valve command/feedback, zone or coil demand, water temperatures/flow where commissioned, and controlled process value. Coverage / timing: mapped control valves with verified actuator scaling and a known served loop/coil. 10-60 s plant loops; 1-5 min zone loops; aggregation: 15-30 min end-stop episodes. Quality / provenance: exclude two-position valves and documented full-open design states; feedback may be command echo rather than true position. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - end-stop persistence plus unmet-demand analysis. Decision interval: rolling 15-30 min. Minimum evidence: healthy valve output and process value for at least 15 min plus active demand/target. Method: detect sustained near-0/near-100% output and determine whether the controlled variable remains outside target. Uncertainty: report whether true position, flow or heat-transfer proof exists. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators distinguish normal saturation from valves or systems that merit a closer check.

Answer boundary

Do not conclude valve failure, blockage or undersizing without physical/flow evidence. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-035

Which pumps are commanded on but do not show the expected independent run, flow, pressure or electrical evidence?**Sensors and telemetry**

Signals / location: pump command, run status/VFD, differential pressure, flow and motor current/power where commissioned. Coverage / timing: mapped pumps with documented proof points and normal startup delay. 5-30 s; aggregation: 2-5 min mismatch episodes. Quality / provenance: proof thresholds must be pump/system specific; shared DP or flow may not prove an individual parallel pump. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M;/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: observation - pump command-to-proof reconciliation. Decision interval: every 1 min while commanded on. Minimum evidence: on-command plus at least one healthy independent proof for longer than the documented startup delay. Method: compare command with run/VFD and available hydraulic/electrical response, grading evidence strength by independence. Uncertainty: state whether the evidence can distinguish one pump in a duty/standby pair. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Allows the operator to focus investigation on commands that are not producing credible system response.

Answer boundary

Do not start, stop, reset or isolate a pump, or declare a mechanical/electrical fault conclusively. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-036

Which VAV or terminal units are at an airflow or actuator limit while their zone demand remains unsatisfied?**Sensors and telemetry**

Signals / location: airflow setpoint/measurement, damper command/feedback, zone temperature, effective target, reheat output and parent AHU pressure/mode. Coverage / timing: mapped VAV/terminal units with commissioned airflow sensing and parent-AHU relationship. 30-60 s; aggregation: 15 min limit episodes. Quality / provenance: airflow sensor zero/balance status and design min/max must be current; terminal demand may be intentionally capped. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M;/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - terminal saturation and unmet-demand analysis. Decision interval: every 5 min during active demand. Minimum evidence: healthy airflow, damper output and zone target/temperature for at least 15 min plus parent AHU context. Method: identify sustained min/max airflow or damper limit with persistent zone error and check upstream pressure/mode evidence. Uncertainty: show whether limit is configured, measured or inferred and whether reheat complicates interpretation. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators determine whether a zone issue is local to the terminal or may originate upstream.

Answer boundary

Do not increase airflow limits, change balancing values or conclude the terminal is undersized. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-037

Which heating or cooling coils are enabled but show little of the expected air- or water-side temperature change?**Sensors and telemetry**

Signals / location: coil enable/valve output, entering/leaving air temperature, entering/leaving water temperature, flow where commissioned, and fan/pump proof. Coverage / timing: coils with verified sensor placement and service mapping. 30-60 s; aggregation: 10-15 min steady-state windows. Quality / provenance: temperature sensors must be plausibly located and healthy; bypass/mixing and low load can legitimately reduce delta-T. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M;/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - coil response screening. Decision interval: after 10 min of stable enabled demand. Minimum evidence: stable enable/demand, healthy entering/leaving temperatures and fan/pump proof for the evaluation window. Method: calculate observed delta-T and trend, then compare with trusted same-mode history or commissioned expectation when available. Uncertainty: show load/flow evidence gaps and avoid using a generic delta-T threshold across different coils. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives operators an evidence-based trigger for heat-transfer investigation rather than relying on command position alone.

Answer boundary

Do not certify coil performance, diagnose fouling/airlock/refrigerant fault, or change valve/plant operation. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-038

Is heat recovery or bypass currently behaving consistently with the approved operating sequence and measured air conditions?**Sensors and telemetry**

Signals / location: heat-recovery enable/speed, bypass damper command/feedback, outdoor/extract/supply temperatures and frost/defrost state where exposed. Coverage / timing: AHUs with documented heat-recovery sequence and commissioned temperature sensors. 30-60 s; aggregation: 10 min operating episodes. Quality / provenance: sequence logic may intentionally bypass recovery for free cooling, frost protection or humidity control. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M;/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - sequence-consistency and temperature-effect screening. Decision interval: every 5 min while AHU is running. Minimum evidence: current approved sequence/mode plus healthy recovery/bypass state and relevant air temperatures. Method: classify expected recovery/bypass state from documented conditions, compare with live state and check for observed temperature effect. Uncertainty: show when sensor uncertainty or undocumented sequence branches prevent a conclusion. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators spot candidate heat-recovery or sequence issues without overriding protective logic.

Answer boundary

Do not force bypass/recovery state or label a life-safety/frost protection action incorrect. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-039

Which fans, pumps, compressors or other HVAC plant items are short-cycling more than their documented operating pattern allows?**Sensors and telemetry**

Signals / location: start/stop events, run proof, stage command, anti-cycle timer state where exposed, demand and fault/reset events. Coverage / timing: assets with reliable event history and a current expected minimum run/off-time or trusted baseline. event-level; aggregation: starts per hour, run/off duration and 24 h summary. Quality / provenance: deduplicate events and separate test/maintenance/restart sequences; compressors may have controller-enforced anti-cycle logic. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M;/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: calculation - cycling-rate and dwell-time analysis. Decision interval: rolling 4 h and 24 h. Minimum evidence: at least 4 h of trustworthy start/stop history plus expected dwell/cycling criteria or stable historical baseline. Method: count starts, calculate dwell distributions and relate cycles to demand/fault/reset episodes. Uncertainty: state criteria source and sample length; unusual does not equal damaging. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators identify assets where unstable demand, controls or faults justify engineering attention.

Answer boundary

Do not reset equipment, change anti-cycle limits or certify that cycling is mechanically harmful. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-040

For this zone problem, what verified AHU, branch, terminal and actuator path currently serves the zone?**Sensors and telemetry**

Signals / location: current asset/service topology identifiers, AHU/branch/terminal states, airflow/valve/damper points and zone environmental signals. Coverage / timing: only links owned by the current asset/BMS topology source; no topology inferred from drawing proximity. topology on change plus 1-5 min states; aggregation: one trace for selected zone/time. Quality / provenance: partial or conflicting service mappings must be exposed; the May 2021 floor plan cannot establish current HVAC service. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS plant/control owner, current asset-register owner, O&M;/commissioning record owner, and CMMS maintenance owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: observation - service-path trace with live-state overlay. Decision interval: on demand during troubleshooting. Minimum evidence: current zone-to-terminal-to-AHU mapping and at least one healthy live state at each available layer. Method: traverse authoritative service links and overlay relevant current states, stopping at missing/conflicting links. Uncertainty: show mapping provenance and gaps separately from telemetry gaps. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets the operator identify the correct upstream equipment to inspect without relying on outdated spatial assumptions.

Answer boundary

Do not invent a service path, assume physical adjacency means connection, or authorise isolation/work from the trace. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-041

What new critical or high-priority HVAC/BMS alarms have appeared during this shift, and which are still active?**Sensors and telemetry**

Signals / location: alarm event stream with point/asset, priority, state transition, acknowledgement state if permitted read-only, and related asset mode. Coverage / timing: authorised HVAC/BMS alarms only; life-safety systems remain governed by their own authoritative interfaces/procedures. event-level; aggregation: shift timeline plus current active set. Quality / provenance: alarm priority mappings and timestamps must be current; connector delay is shown explicitly. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - new-alarm extraction and current-state reconciliation. Decision interval: continuous/event-driven with shift summary. Minimum evidence: authoritative alarm event/state feed for the shift and verified asset mapping. Method: identify first occurrence, repeats, current state and affected asset without deducing severity beyond configured priority. Uncertainty: show retrieval delay, duplicate-event handling and any gap in event coverage. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives the operator a concise shift alarm picture without replacing the BMS alarm workflow.

Answer boundary

Do not acknowledge, suppress or clear alarms, reinterpret life-safety severity, or promise that an inactive alarm is resolved. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-042

Which alarms are chattering - repeatedly clearing and returning - rather than occurring as a single stable event?**Sensors and telemetry**

Signals / location: alarm on/off transitions, point value around each transition, alarm threshold/deadband where authorised, and controller restart events. Coverage / timing: alarms with reliable event history and known state semantics. event-level with associated native point samples; aggregation: transitions per hour and episode duration. Quality / provenance: remove duplicate notifications but preserve genuine state changes; communication loss can mimic alarm cycling. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: calculation - alarm-chatter detection. Decision interval: rolling 4 h and 24 h. Minimum evidence: at least 6 transitions or 2 h of complete event history plus threshold/value context where available. Method: count on/off cycles, measure dwell times and compare point values with configured threshold/deadband. Uncertainty: show whether telemetry gaps or restarts could explain the pattern. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators target nuisance-alarm investigation while retaining the formal alarm process.

Answer boundary

Do not change alarm thresholds/deadbands, suppress nuisance alarms or conclude the underlying plant is unstable. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-043

Which active HVAC alarms have persisted beyond the current response target or beyond the operating state that normally explains them?**Sensors and telemetry**

Signals / location: alarm active-since time, priority, asset state, related schedule/mode and authorised response-target metadata where available. Coverage / timing: active authorised HVAC/BMS alarms with current response rules. event-level plus 1-5 min asset context; aggregation: alarm age/status list. Quality / provenance: response targets may differ by priority, time and service; missing target is not replaced by a guessed SLA. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - alarm-age and context persistence analysis. Decision interval: every 5 min for active high-priority alarms. Minimum evidence: active-since timestamp, current state, authorised response target or documented expected-mode condition. Method: calculate alarm age and determine whether the explaining mode/condition has ended while alarm persists. Uncertainty: show target source and pause/exception periods if formally defined. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Supports escalation of long-lived alarms with the context needed to avoid false urgency.

Answer boundary

Do not declare a breach, assign responsibility, or clear/acknowledge the alarm. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-044

Which BMS sensors currently look flatlined, stale, missing, implausible or subject to repeated jumps?**Sensors and telemetry**

Signals / location: raw sensor values, quality/status flags, last-seen timestamps, expected engineering range and controller communication state. Coverage / timing: commissioned environmental and plant sensors the operator is permitted to view. native point sampling; aggregation: quality features over 1 h and 24 h. Quality / provenance: expected ranges must be point-specific; a stable value can be legitimate and is not automatically a flatline fault. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - multi-rule sensor-quality screening. Decision interval: every 15 min with immediate stale/communication flags. Minimum evidence: minimum 1 h recent data plus point metadata; 24 h preferred for flatline/jump context. Method: test freshness, missingness, rate-of-change, impossible range and low-variance patterns using point-specific tolerances. Uncertainty: report which rule triggered and its threshold; statistical anomaly is not proof of sensor failure. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets operators separate likely telemetry-quality problems from genuine HVAC behaviour before diagnosis.

Answer boundary

Do not auto-substitute another sensor, calibrate the point or declare the physical condition normal because a sensor is bad. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-045

Which paired, redundant or independently comparable sensors disagree beyond their approved combined tolerance?**Sensors and telemetry**

Signals / location: paired sensor readings, timestamps, calibration dates, accuracy class and known placement/measurement role. Coverage / timing: only sensor pairs explicitly documented as redundant/comparable. native sampling aligned to the slower sensor; aggregation: 15 min paired bias and 24 h distribution. Quality / provenance: do not compare sensors that measure different locations or media merely because names are similar. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - paired-sensor agreement test. Decision interval: every 15 min and daily trend. Minimum evidence: two healthy comparable sensors, at least 30 aligned samples and approved tolerance or stated accuracy. Method: calculate bias and spread against combined tolerance, tracking whether disagreement is persistent or operating-state dependent. Uncertainty: show calibration age and pair-definition provenance. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides a defensible shortlist for calibration or wiring/mapping checks.

Answer boundary

Do not choose a "correct" sensor without a trusted reference or authorised calibration check. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-046

Which temperature sensors show evidence of slow drift relative to a trusted reference or stable peer relationship?**Sensors and telemetry**

Signals / location: sensor temperature history, trusted reference/peer values, calibration/maintenance events, mode and relevant process conditions. Coverage / timing: sensors with a documented reference or historically stable comparable relationship. 1-5 min; aggregation: daily bias estimate over 2-8 weeks. Quality / provenance: seasonal/process changes can alter peer relationships; comparisons across remapping or sensor replacement are invalid. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - robust drift estimation. Decision interval: daily or weekly, not for immediate control. Minimum evidence: at least 10 comparable days or a documented reference-check series with stable mapping. Method: estimate median bias trend using matched operating periods and break the series at calibration/replacement/configuration events. Uncertainty: show trend uncertainty and reference stability; low evidence -> no drift claim. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps maintenance teams prioritise sensors for verification before drift distorts control decisions.

Answer boundary

Do not recalibrate automatically, alter setpoints to compensate, or label drift as calibration failure without a competent check. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-047

Which actuator feedback points may be stuck, inverted, badly scaled or mapped to the wrong device?**Sensors and telemetry**

Signals / location: actuator command, position feedback, end switches, related flow/temperature/pressure response and recent commanded movement events. Coverage / timing: actuators with commissioned feedback and a verified served process. 10-60 s during movement; aggregation: movement-response episodes. Quality / provenance: command echoes are not independent feedback; polarity/scaling and actuator travel time must be known. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - actuator signal-consistency diagnostics. Decision interval: after each significant command change and daily assessment. Minimum evidence: at least 3 distinct commanded positions or one full movement event with feedback and process-response context. Method: test command-feedback slope, direction, range and physical-response consistency; flag alternative explanations rather than one diagnosis. Uncertainty: show whether evidence supports stuck, inverted, scaled, mapped-wrong or simply inconclusive. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives controls and maintenance staff a focused diagnostic hypothesis instead of a vague "bad actuator" alert.

Answer boundary

Do not move the actuator for testing, rewrite mapping/scaling or declare a mechanical defect conclusively. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-048

Which BMS points recently changed engineering units, scaling, calibration constants, mapping or alarm configuration?**Sensors and telemetry**

Signals / location: point-database configuration history, configuration versions, calibration parameters, engineering units, scaling and alarm metadata. Coverage / timing: authorised BMS configuration objects and change-control records. event/version based; aggregation: changes by shift/day/week. Quality / provenance: configuration-history availability varies by platform; absence of a configuration event is not proof that no external change occurred. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - configuration-delta extraction. Decision interval: after approved change windows and at shift assessment. Minimum evidence: at least two trusted configuration versions or an authoritative configuration event with actor/role permissions. Method: diff selected point semantics/settings and link to approved change record where available. Uncertainty: show effective time, source system and any gaps in configuration history. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps explain sudden telemetry or alarm behaviour that may result from configuration rather than physical plant change.

Answer boundary

Do not reveal unnecessary personal identity, reverse changes, or certify that undocumented changes are unauthorised. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-049

Which current HVAC alarms and abnormal signals could share the same upstream plant cause?**Sensors and telemetry**

Signals / location: time-stamped alarms, plant/zone process values, command/proof states and verified service topology. Coverage / timing: assets connected by current authoritative HVAC service relationships only. event-level plus 1-5 min process data; aggregation: correlated episodes over 30-120 min. Quality / provenance: correlation requires aligned clocks and current topology; shared timing alone does not prove common cause. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - topology-constrained event clustering. Decision interval: on demand for active incident and every 15 min during it. Minimum evidence: at least two related abnormal signals within a verified service path and sufficient pre/post event coverage. Method: cluster events by time and upstream/downstream topology, then rank candidate common sources while preserving alternatives. Uncertainty: present hypotheses with supporting/contradicting evidence, not a single root-cause claim. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps the operator avoid treating many downstream alarms as unrelated when one upstream problem may explain them.

Answer boundary

Do not declare root cause, bypass alarms or direct unsafe intervention from correlation alone. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-050

For this suspected BMS/HVAC fault, is the evidence strong enough for a diagnosis, or should we stop at "unknown" and verify more?**Sensors and telemetry**

Signals / location: all relevant point health, process values, commands/feedback, alarms, topology, configuration and maintenance context for the selected fault. Coverage / timing: only the verified affected service path and authorised time window. native timing; aggregation: diagnostic evidence matrix. Quality / provenance: absence of a sensor, run proof or configuration record is a diagnostic limitation rather than negative evidence. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS alarm/configuration owner, point-database/change-control owner, calibration/maintenance owner, and CMMS owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - diagnostic evidence sufficiency and contradiction check. Decision interval: on demand before escalation or work instruction. Minimum evidence: at least one trustworthy symptom plus the minimum independent evidence required by the candidate diagnostic rule. Method: evaluate candidate explanations against supporting, contradicting and missing evidence; stop when discrimination is inadequate. Uncertainty: show exactly what additional observation/check would most reduce uncertainty. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Prevents weak telemetry from becoming a confident fault story and points the operator to the next useful verification.

Answer boundary

Do not present an unverified diagnosis as fact, authorise maintenance action, or certify equipment safety. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-051

Which after-hours HVAC extensions have passed their verified end time and are ready for authorised operator assessment?**Sensors and telemetry**

Signals / location: extension state/expiry, affected schedule and plant/zone enable, current mode and approved exception/maintenance windows. Coverage / timing: all active extension mechanisms with verified scope and expiry semantics. event-level plus 1-5 min; aggregation: expired-active list. Quality / provenance: differentiate passed expiry from missing expiry and account for authorised extension renewal/change events. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - expired-extension candidate detection. Decision interval: every 5 min while extensions are active. Minimum evidence: active extension, trustworthy expiry/end condition and current authorised exception context. Method: flag extensions that remain active beyond verified end condition, excluding current authorised maintenance/emergency exceptions. Uncertainty: show expiry provenance and delay since end time. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Supports prompt assessment of temporary runtime that may otherwise persist unnoticed.

Answer boundary

Do not cancel the extension, return the system to normal mode or decide that continued operation is unnecessary. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-052

Where is optimum start consistently bringing HVAC on materially earlier than needed to meet the approved occupied target?**Sensors and telemetry**

Signals / location: optimum-start enable time, occupied start, zone temperature/target, plant proof and outdoor temperature where verified. Coverage / timing: zones/systems where optimum start is actually configured and historical recovery evidence is trustworthy. 1-5 min; aggregation: per start episode over 2-6 weeks. Quality / provenance: exclude unusual weather, maintenance, closures and days where occupied target was not achieved. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - start-lead-time performance comparison. Decision interval: weekly assessment, not real-time control. Minimum evidence: at least 10 comparable successful start episodes or a documented commissioned baseline. Method: measure lead time and time target is first achieved, then identify repeated excess margin under comparable conditions. Uncertainty: show weather-match quality and sample distribution; do not convert the estimate directly into a new schedule. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides evidence for a controls engineer to assess whether early-start logic is overly conservative.

Answer boundary

Do not change optimum-start parameters/schedule or promise energy savings. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-053

Where is preheat or precool repeatedly failing to reach the approved target by occupied start?**Sensors and telemetry**

Signals / location: preheat/precool start, occupied start, zone/air temperatures, target band, plant enable/proof and outdoor condition. Coverage / timing: systems with documented preconditioning strategy and reliable recovery history. 1-5 min; aggregation: per start episode and weekly failure rate. Quality / provenance: exclude days with approved plant outages or strategy changes; verify target applies to that zone/mode. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - preconditioning success-rate analysis. Decision interval: each occupied start plus weekly assessment. Minimum evidence: at least 5 recent comparable start episodes with complete timing and temperature evidence. Method: calculate target-at-start success, shortfall and recovery trajectory, then associate failures with plant availability and weather. Uncertainty: show sample size and evidence gaps; failure to meet target does not establish cause. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps prioritise whether the issue is scheduling, plant availability, controls or a need for engineering assessment.

Answer boundary

Do not advance schedules, raise setpoints or declare plant capacity inadequate. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-054

Which zones show persistent simultaneous heating and cooling patterns strong enough to justify a controls-engineering assessment?**Sensors and telemetry**

Signals / location: heating/cooling outputs, valve/damper/reheat feedback where available, zone temperature/targets, airflow and operating mode. Coverage / timing: zones with verified independent heating/cooling signals and documented sequence exceptions. 1 min; aggregation: overlap minutes per hour/day over 1-4 weeks. Quality / provenance: exclude dehumidification reheat, frost/defrost, changeover transients and documented minimum reheat strategies. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - persistent overlap opportunity screening. Decision interval: daily/weekly assessment. Minimum evidence: at least 5 valid occupied days with 80% coverage and verified sequence context. Method: quantify overlap magnitude/duration, associated zone error and repeatability, ranking only after known exceptions are removed. Uncertainty: show whether outputs are commands or physical heat/cool proof and avoid energy-savings claims without metering. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Creates a defensible shortlist for controls optimisation rather than chasing brief or intentional overlap.

Answer boundary

Do not retune loops, change deadbands or label the pattern waste without engineering assessment. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-055

Which AHU pressure- or supply-air-temperature reset loops stay at an approved limit while downstream demand suggests headroom or unmet need?**Sensors and telemetry**

Signals / location: reset setpoint, process value, controller output, VFD speed or SAT control output, terminal damper/demand summaries and mode. Coverage / timing: AHUs with documented reset strategy and verified downstream demand aggregation. 10-60 s plant plus 1 min terminal summary; aggregation: 15 min limit episodes and daily duration. Quality / provenance: downstream demand must not be inferred from incomplete terminal coverage; exclude manual/test modes. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - reset-loop limit and downstream-demand screening. Decision interval: every 15 min with weekly trend. Minimum evidence: healthy reset/process/output plus representative downstream demand for at least 30 min. Method: measure time at high/low reset limits and compare with documented downstream demand indicator/sequence conditions. Uncertainty: show terminal coverage percentage and whether limits are configured safeguards. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps identify reset strategies that deserve tuning assessment while respecting incomplete downstream evidence.

Answer boundary

Do not change reset parameters, widen limits or claim the current strategy is inefficient. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-056

In commissioned demand-controlled ventilation areas, where does ventilation demand stay high while CO2 remains persistently low?**Sensors and telemetry**

Signals / location: zone/return CO2, DCV demand signal, outdoor-air damper/airflow, fan speed and occupancy mode; anonymous zone counts only if already authorised. Coverage / timing: only areas with documented DCV sequence and healthy commissioned CO2/airflow evidence. 1-5 min; aggregation: 30 min episodes and daily duration. Quality / provenance: low CO2 alone is not proof of over-ventilation; minimum ventilation, purge, pressure and other sequences may govern. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - DCV consistency/opportunity screening. Decision interval: daily/weekly assessment. Minimum evidence: at least 5 comparable occupied days or 8 h of valid DCV operation with current sequence documentation. Method: identify periods where CO2 is persistently low while documented DCV demand/airflow is high, then remove known minimum/exception modes. Uncertainty: show airflow evidence strength and do not estimate savings without validated fan/thermal energy data. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides a bounded candidate list for checking DCV sequence, sensors or minimum-air settings.

Answer boundary

Do not reduce outdoor air, change DCV thresholds or certify ventilation adequacy. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-057

Which recurring after-hours HVAC runtime has no matching approved schedule, extension, maintenance window or protective operating mode?**Sensors and telemetry**

Signals / location: plant run proof, schedule, extensions, maintenance/test windows, frost/protection modes and relevant outdoor conditions. Coverage / timing: mapped HVAC assets with reliable runtime proof and authoritative exception records. 1-5 min; aggregation: 15 min runtime episodes over 2-4 weeks. Quality / provenance: unrecorded operational instructions remain possible; absence of a matching record is not proof of unauthorised operation. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - exception-aware after-hours runtime screening. Decision interval: daily and weekly assessment. Minimum evidence: at least 10 days of 80% runtime coverage plus current schedules/exceptions for the same period. Method: subtract all verified approved windows/protective modes from run intervals and rank repeated unexplained remainder. Uncertainty: show missing exception-record coverage and metering absence; avoid claiming energy waste amount without energy data. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators find repeatable unexplained runtime for schedule, controls or plant investigation.

Answer boundary

Do not stop plant, remove protective logic or label runtime unauthorised. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-058

Which control loops show repeatable oscillation, overshoot or hunting that is strong enough to justify a tuning assessment?**Sensors and telemetry**

Signals / location: setpoint, process value, controller output, actuator feedback and mode at a sampling rate appropriate to the loop. Coverage / timing: commissioned loops with enough temporal resolution and no manual/test state. 5-30 s fast loops; 30-60 s slower plant loops; 1-5 min only for genuinely slow zone loops; aggregation: oscillation episodes and frequency features. Quality / provenance: under-sampling can create false oscillation; plant staging and setpoint changes must be separated from self-sustained cycling. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - oscillation/overshoot screening. Decision interval: per stable-control episode and weekly assessment. Minimum evidence: at least 6 apparent cycles or 30 min of stable setpoint/mode with sufficient sampling. Method: estimate oscillation period/amplitude, output-process phase behaviour and overshoot following known setpoint steps. Uncertainty: show sampling adequacy and alternative explanations such as staging or sensor quantisation. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives controls staff evidence-rich candidates for tuning instead of relying on visual impressions.

Answer boundary

Do not retune PID parameters, force step tests or declare loop instability without a controls engineer. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-059

Which possible HVAC control opportunities have enough verified evidence for a bounded engineering recommendation, and which should be deferred?

Sensors and telemetry

Signals / location: relevant schedules, setpoints, environmental/plant telemetry, meter data where applicable, point health and current constraints/change records. Coverage / timing: only systems with verified mapping and evidence coverage appropriate to the proposed assessment. native timing by issue; aggregation: evidence package per candidate. Quality / provenance: a candidate with missing cost, safety, sequence or metering context is marked not ready rather than filled with assumptions. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - recommendation-readiness classification. Decision interval: weekly or before an optimisation assessment. Minimum evidence: repeatable issue evidence, current approved strategy, current maintenance/safety constraints and enough data to evaluate the claimed mechanism. Method: classify each candidate as evidence-ready, needs verification, or insufficient; state the smallest next evidence action. Uncertainty: separate confidence in the observed pattern from confidence in any expected benefit. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps the operator bring only well-supported opportunities to controls/energy assessment and park speculative ones.

Answer boundary

Do not authorise implementation, promise savings/comfort implementation or bypass engineering/change-control assessment. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-060

If we can investigate only one BMS/HVAC control issue this shift, which verified issue has the strongest case for attention?

Sensors and telemetry

Signals / location: active alarms, persistent deviations, unresolved command/proof mismatches, override age, affected service scope and evidence-quality indicators. Coverage / timing: current authorised operational evidence across mapped HVAC systems. 1-5 min plus event data; aggregation: current prioritisation snapshot. Quality / provenance: priority logic must respect formal alarm/safety procedures and cannot downgrade mandatory responses. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS controls and operational-strategy owner, energy-management owner, current maintenance constraint owner, and relevant commissioning-baseline owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - bounded issue prioritisation. Decision interval: on request during shift triage. Minimum evidence: at least two candidate issues with verified persistence/impact evidence and current formal priority/context. Method: rank after mandatory safety/alarm obligations by persistence, service scope, evidence strength and reversibility of investigation - not by guessed cost. Uncertainty: show scoring factors and ties; do not hide candidates with missing data. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides a transparent triage aid when operator attention is limited.

Answer boundary

Do not override formal priorities, tell staff to ignore alarms, or authorise corrective action. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-061

What unresolved BMS/HVAC issues need to be handed over to the next shift, with current state and evidence age?**Sensors and telemetry**

Signals / location: active alarms, overrides, abnormal diagnostics, plant mode, work-order status and authorised operator notes. Coverage / timing: current shift scope; personal details reduced to role/team where identity is not operationally necessary. event-level plus current snapshot; aggregation: one handover item per issue. Quality / provenance: notes are context, not authoritative replacement for alarm/CMMS/configuration state; show retrieval age. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - shift-handover synthesis. Decision interval: at handover and on demand. Minimum evidence: current authoritative state for each open issue plus at least one recent observation and any linked work-order/change record. Method: summarise what happened, current state, verified constraints, pending authorised action and evidence gaps. Uncertainty: distinguish operator note from source-of-record fact and show last observation time. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Reduces loss of context between shifts and makes unresolved risks/unknowns visible.

Answer boundary

Do not assign responsibility, close work orders/alarms or imply the next shift has accepted an action. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-062

Which sensors, actuators or plant items should be physically checked before we rely on their BMS telemetry for this diagnosis?**Sensors and telemetry**

Signals / location: point-quality flags, command/feedback consistency, redundant-sensor comparison, recent maintenance/calibration and communication health. Coverage / timing: the selected diagnostic scope and its verified service path. native timing; aggregation: verification shortlist. Quality / provenance: a diagnostic anomaly is only a reason to check, not evidence that the physical device is defective. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - physical-verification prioritisation. Decision interval: on demand before maintenance escalation. Minimum evidence: at least one independent reason the telemetry may be unreliable or contradictory. Method: rank checks by diagnostic leverage, accessibility information from current authorised records and whether independent verification exists. Uncertainty: show why each check is suggested and what result would resolve uncertainty. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps maintenance effort focus on the few physical checks most likely to clarify the fault.

Answer boundary

Do not instruct unsafe access, infer access/hazards from the May 2021 floor plan or substitute for permit-to-work procedures. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-063

Which newly commissioned, replaced or remapped BMS points have not yet demonstrated stable and plausible behaviour?**Sensors and telemetry**

Signals / location: point commissioning/change date, raw value/status, engineering units, expected range, mapping, related peer/process signals and fault flags. Coverage / timing: points changed within the authorised commissioning window. native timing; aggregation: post-change stability over hours/days appropriate to the point. Quality / provenance: new points may legitimately see limited operating range initially; use point-specific commissioning acceptance evidence. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - post-change telemetry validation. Decision interval: daily until acceptance or for the defined observation period. Minimum evidence: authoritative change/commissioning record plus sufficient post-change samples across at least one relevant operating state. Method: check freshness, range, variability, directionality and process consistency, separating untested operating states. Uncertainty: show which behaviours have and have not yet been observed. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Shows commissioning staff which points still need evidence before being trusted operationally.

Answer boundary

Do not certify commissioning acceptance or rewrite the point mapping/configuration. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-064

Which open CMMS work orders match the BMS symptoms we are seeing now, and which symptoms have no current work-order context?**Sensors and telemetry**

Signals / location: current BMS alarms/diagnostics and asset identifiers; no contractor personal data beyond role/organisation where needed. Coverage / timing: verified asset IDs shared between BMS and CMMS. BMS event/current state plus CMMS retrieval; aggregation: issue-to-work-order links. Quality / provenance: asset identity must be reconciled explicitly; free-text similarity alone is weak evidence. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - asset/time constrained BMS-CMMS correlation. Decision interval: on demand and every 15-30 min during an incident. Minimum evidence: current BMS symptom with verified asset ID and permitted open CMMS records. Method: join exact asset IDs first, then use time/symptom text only as secondary context; show unmatched items separately. Uncertainty: report match basis and CMMS staleness; no match is not proof no maintenance is planned. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Prevents duplicate troubleshooting and gives operators the maintenance context already known about an asset.

Answer boundary

Do not create/close/reassign work orders or disclose contractor details beyond operational need. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-065

After recent maintenance, did the affected HVAC system return to the documented expected BMS behaviour?**Sensors and telemetry**

Signals / location: maintenance completion/time, relevant commands/feedback, process values, alarms and pre/post operating state. Coverage / timing: the maintained asset and verified downstream/upstream service context. native timing; aggregation: pre/post windows matched by operating mode. Quality / provenance: maintenance completion in CMMS is not itself proof of physical performance; pre/post conditions must be comparable. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - post-maintenance response verification. Decision interval: after the first comparable operating episode and again after 24 h where useful. Minimum evidence: authoritative maintenance record plus healthy pre/post telemetry under at least one comparable operating state. Method: compare defined expected signals/response with pre-maintenance symptom and commissioned/trusted normal range. Uncertainty: show changes in mode/load and whether all acceptance conditions were observable. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets operators see whether the operational symptom actually disappeared after maintenance.

Answer boundary

Do not certify maintenance completion, commissioning acceptance or equipment safety. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-066

Which valves, dampers, sensors or feedback points have accumulated enough questionable behaviour to justify functional verification?**Sensors and telemetry**

Signals / location: repeated command/feedback anomalies, sensor-quality flags, disagreement episodes, work-order history and calibration/verification dates. Coverage / timing: commissioned devices with a current asset/point identity. events plus native telemetry; aggregation: 30-90 day issue history. Quality / provenance: do not turn one rare anomaly into a maintenance requirement; consider known outages and changed mappings. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - verification-due candidate scoring. Decision interval: weekly/monthly planning. Minimum evidence: at least 3 independent anomaly episodes, or one persistent high-impact contradiction, plus current maintenance history. Method: combine recurrence, diagnostic impact and elapsed time since last verification without replacing statutory/PPM schedules. Uncertainty: show evidence basis and whether the candidate is already covered by planned work. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Supports evidence-informed maintenance planning alongside - not instead of - formal maintenance duties.

Answer boundary

Do not alter statutory/PPM frequency, authorise maintenance or certify an asset is out of tolerance. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-067

What evidence package should accompany a contractor or specialist call-out for this HVAC fault?**Sensors and telemetry**

Signals / location: selected alarm timeline, commands/feedback, process trends, current mode, asset ID, recent changes and work-order reference. Coverage / timing: only data necessary for the task and permitted for external sharing; redact credentials, personal data and unrelated research/room information. native timing; aggregation: bounded incident packet. Quality / provenance: external permissions may be narrower than operator permissions; sharing must use approved channels. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - minimal evidence-package assembly. Decision interval: on demand before authorised call-out. Minimum evidence: verified asset identity, fault time window, current state and enough diagnostic evidence to reproduce the symptom context. Method: select smallest relevant trend/event set with timestamps, units, provenance, point-health caveats and current known constraints. Uncertainty: show what is withheld or unavailable rather than silently omitting critical context. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives specialists useful diagnostic context while reducing repeated data gathering and oversharing.

Answer boundary

Do not contact/engage a contractor, disclose credentials or unnecessary personal/sensitive building data, or issue a method of work. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-068

What authorised BMS changes took effect today, and which live points confirm the intended state is now active?**Sensors and telemetry**

Signals / location: change-control records, configuration versions, schedule/setpoint/mapping changes, controller readback and affected asset states. Coverage / timing: approved changes within the current day/shift and their stated target scope. event/version based plus 1-5 min verification; aggregation: change-to-live-state record. Quality / provenance: separate approved intent, deployment event and effective controller state; each can have a different time. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - change-effect verification. Decision interval: after each change window and at handover. Minimum evidence: approved change record, target scope/version and healthy live readback/behaviour from affected points. Method: trace approval/effective time to deployed version and observed controller state, flagging targets with missing verification. Uncertainty: show incomplete verification and any later superseding change. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides the next operator with a factual picture of what changed and whether it appears to have taken effect.

Answer boundary

Do not roll back/reapply changes, certify commissioning or reveal unnecessary individual actor details. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-069

Which temporary overrides, isolations, test modes or engineering states still need authorised assessment before this shift ends?**Sensors and telemetry**

Signals / location: override/test/engineering mode flags, isolation status where read-only authorised, expiry, work-order/permit references and affected service state. Coverage / timing: only states visible under the operator's authorised scope; safety isolations remain governed by permit/source-of-record systems. event-level plus current state; aggregation: end-of-shift temporary-state list. Quality / provenance: BMS indication may not be the authoritative isolation record; permit-to-work records take precedence. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - temporary-state reconciliation. Decision interval: 30-60 min before handover and on demand. Minimum evidence: current temporary-state telemetry plus authoritative permit/work/change record where applicable. Method: list each state, age, expiry/reference and affected service, highlighting missing source-of-record confirmation. Uncertainty: label BMS-only indication separately from verified isolation/permit status. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Reduces the chance that temporary engineering states are forgotten across shift boundaries.

Answer boundary

Do not remove overrides, release isolation, close permits or declare equipment safe to return to service. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-070

For this recurring zone or plant problem, what is the concise handover from first verified symptom to current state?**Sensors and telemetry**

Signals / location: time-stamped alarms, environmental/process trends, commands/feedback, overrides, work orders, maintenance/change events and authorised notes. Coverage / timing: the verified affected service path and requested incident period only. event/native samples; aggregation: chronological incident timeline. Quality / provenance: preserve original timestamps and source labels; operator notes must not overwrite source-of-record events. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS and CMMS maintenance owners, permit/change-control owner, commissioning owner, contractor record owner, and shift-log owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: lookup - incident chronology synthesis. Decision interval: on demand for handover/escalation. Minimum evidence: a verified first symptom, current state, and sufficient events to explain major transitions; gaps remain explicit. Method: order key facts, actions already recorded, response changes and unresolved evidence, separating observation from hypothesis. Uncertainty: show gaps, clock conflicts and superseded interpretations. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Lets the next operator or engineer pick up a recurring issue without rereading hours of raw trends and notes.

Answer boundary

Do not assign cause or responsibility, close the incident or authorise corrective action. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-071

Which HVAC systems are contributing most to current electrical demand where suitable metering is actually available?**Sensors and telemetry**

Signals / location: verified HVAC submeters or equipment power, plant run state and meter quality/interval data. Coverage / timing: only metered circuits/assets with a current electrical-to-HVAC mapping; do not allocate shared/unmetered load by guesswork. 1-15 min depending on meter; aggregation: current demand and 15 min demand interval. Quality / provenance: meter time base, units, multiplier and circuit scope must be verified; command state is not a substitute for power measurement. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - current HVAC demand attribution. Decision interval: each utility demand interval, refreshed when new meter data arrives. Minimum evidence: healthy meter data covering the current interval and verified circuit/asset mapping. Method: rank measured HVAC loads and reconcile submeter sum with known parent meter only where scopes are compatible. Uncertainty: show unmetered/shared load and meter latency; do not estimate asset demand from nameplate alone. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives operators a factual view of measured HVAC demand contributors during high-load periods.

Answer boundary

Do not shed loads, change operation or claim whole-building demand attribution where metering is incomplete. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-072

Are we likely to cross an approved peak-demand threshold in the current demand interval, and which HVAC loads are formally classed as essential, constrained or discretionary?**Sensors and telemetry**

Signals / location: current demand meter, interval forecast input, HVAC submeter/run states and read-only load-classification metadata. Coverage / timing: metered demand scope and only assets covered by the current authorised demand-response/resilience policy. 1-5 min meter if available; aggregation: current/forecast utility interval. Quality / provenance: load class must come from current authorised policy; missing classification stays unknown and forecast uncertainty must be visible. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: forecast - short-horizon peak-demand projection plus authorised load-class lookup. Decision interval: each demand interval, refreshed on new meter samples. Minimum evidence: a healthy interval demand meter, the approved threshold/contract rule, at least three recent aligned samples, and current authorised load classifications. Method: conservatively project end-of-interval demand; compare the projection and uncertainty band with the approved threshold; list load classes separately; never propose automatic shedding. Uncertainty: show forecast horizon, meter latency, recent volatility and a validated forecast error band if available; otherwise label the projection low-confidence. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps the operator prepare for an authorised demand response without improvising which services may be changed.

Answer boundary

Do not shed/stop loads, reclassify essential services or guarantee the peak will be avoided. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-073

Which HVAC systems ran materially longer than their approved operating windows yesterday?**Sensors and telemetry**

Signals / location: run proof/runtime counters, schedule/exception/maintenance windows and protective-mode events. Coverage / timing: mapped HVAC assets with reliable runtime evidence. 1-5 min or runtime counter; aggregation: daily runtime by asset/system. Quality / provenance: exclude approved extensions, maintenance, frost/protection and emergency operation; runtime counters must be rollover-safe. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: calculation - exception-aware daily runtime variance. Decision interval: daily assessment. Minimum evidence: at least 90% valid runtime coverage plus authoritative approved windows for the day. Method: calculate actual versus approved runtime after subtracting verified exceptions and rank only persistent excess. Uncertainty: show data completeness and do not convert excess hours to energy without metering or validated power models. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides a clear starting point for investigating repeatable schedule or control-runtime problems.

Answer boundary

Do not stop plant, alter schedules or label excess runtime as waste/unauthorised. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-074

How does today's measured HVAC energy compare with genuinely comparable weather-normalised days?**Sensors and telemetry**

Signals / location: HVAC energy meters, outdoor temperature/weather, operating hours/mode and major approved closure/maintenance flags. Coverage / timing: metered HVAC scope with stable meter boundaries over the comparison period. 15-30 min meter and weather; aggregation: daily energy with degree-hour or matched-day features. Quality / provenance: meter replacements, boundary changes, exceptional occupancy/operations and missing weather can invalidate comparison. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - weather/operation normalised energy comparison. Decision interval: daily after sufficient data are available. Minimum evidence: at least 20 comparable historical days or a validated baseline model with current meter boundary. Method: use matched-day or simple regression with weather and operating hours, hold out recent data when validating baseline error. Uncertainty: report prediction interval and baseline fit; correlation is not proof of why energy changed. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators see whether energy use is unusual after accounting for major weather and operating-time effects.

Answer boundary

Do not certify energy performance, claim savings or penalise operational teams from model residuals alone. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-075

Which metered HVAC systems are using substantial energy while showing weak delivered heating or cooling effect?**Sensors and telemetry**

Signals / location: equipment/system energy or power plus appropriate delivered-effect signals such as air/water flow and entering/leaving temperature. Coverage / timing: only systems where energy and thermal-delivery measurements refer to the same verified boundary. 1-15 min; aggregation: 15-60 min steady operating windows. Quality / provenance: do not compare unmatched electrical and thermal boundaries; low delta-T can be normal at low load or high flow. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: estimate - energy-to-delivered-effect screening. Decision interval: daily/weekly assessment under stable operation. Minimum evidence: healthy energy plus flow and temperature evidence for at least 30 min in comparable mode. Method: estimate thermal effect where measurements support it, or use a bounded proxy; compare only against same-system trusted history. Uncertainty: propagate meter/sensor uncertainty and state when thermal output cannot be calculated defensibly. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Identifies systems where measured energy and delivered effect have diverged enough to justify engineering assessment.

Answer boundary

Do not certify efficiency/COP, diagnose equipment degradation or promise energy savings without appropriate engineering metering. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-076

Are seasonal changeover settings and active HVAC modes aligned with the current approved seasonal strategy and recent outdoor conditions?**Sensors and telemetry**

Signals / location: season/changeover state, heating/cooling plant availability, reset schedules, outdoor temperature and relevant lockout thresholds. Coverage / timing: systems governed by a documented current seasonal/changeover strategy. 5-15 min; aggregation: current state plus 7 day weather/context. Quality / provenance: outdoor sensor and weather feed must be verified; season strategy may be calendar-, temperature- or manually governed. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - seasonal-strategy consistency check. Decision interval: daily and after approved changeover events. Minimum evidence: current approved strategy/version, healthy outdoor condition and live plant/mode/changeover state. Method: evaluate whether active state follows the documented trigger/lockout logic, showing any manual authorised exception. Uncertainty: show hysteresis and weather-data uncertainty; recent weather alone does not define the authorised season. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Helps operators catch configuration drift around seasonal transitions without automating the changeover.

Answer boundary

Do not switch seasonal mode, enable/disable plant or redefine changeover thresholds. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-077

Which HVAC assets show degraded seasonal performance compared with their own commissioned or otherwise trusted baseline?**Sensors and telemetry**

Signals / location: asset energy/power, flow, temperatures, run state, load proxy and outdoor conditions where relevant. Coverage / timing: assets with stable metering/sensor boundaries and a documented commissioned/trusted baseline. 1-15 min; aggregation: weekly seasonal performance features. Quality / provenance: baseline must be versioned and broken when major equipment, controls or meter boundaries change. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - baseline residual/trend analysis. Decision interval: weekly/monthly seasonal assessment. Minimum evidence: at least 4 weeks current data across relevant load range plus a validated baseline of sufficient historical coverage. Method: compare current performance features with baseline conditioned on load/weather, checking for persistent residual shift. Uncertainty: show baseline error, change points and data-boundary continuity; degradation is a screening signal, not a diagnosis. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Provides early evidence for engineering investigation before poor seasonal performance becomes obvious operationally.

Answer boundary

Do not certify degradation, remaining life or maintenance need from analytics alone. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-078

During a power, network or BMS disruption, which HVAC evidence is still trustworthy, delayed, cached or unavailable?**Sensors and telemetry**

Signals / location: controller heartbeat, gateway/network status, local trend availability, meter continuity, UPS-backed telemetry status and timestamp/source metadata. Coverage / timing: authorised BMS/telemetry architecture only; do not infer physical service continuity from data visibility alone. event-level; aggregation: current data-availability matrix. Quality / provenance: cached values must be labelled with original observed time; gateway health is not proof of field-device health. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: comparison - disruption-aware evidence availability classification. Decision interval: immediate during disruption and recovery. Minimum evidence: last-known health/heartbeat, timestamp provenance and known data paths for the affected systems. Method: classify each evidence stream as live, locally buffered, delayed, stale, unavailable or unknown and update on recovery. Uncertainty: show age and path dependency; absence of telemetry cannot establish plant state. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Prevents operators from mistaking stale dashboards for live plant state during communications or power incidents.

Answer boundary

Do not claim HVAC services are operating safely, restart systems or substitute for resilience/emergency procedures. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-079

Which critical HVAC services depend on a single telemetry or control-visibility path with no verified alternative evidence?**Sensors and telemetry**

Signals / location: current BMS/network architecture, controller/gateway dependencies, point-source mapping, meter/proof redundancy and resilience classification. Coverage / timing: only current authorised architecture records; no assumption from legacy drawings or the May 2021 floor plan. Architecture on change plus current health; aggregation: dependency graph with live-health overlay. Quality / provenance: control redundancy and telemetry redundancy are different; an alternate display point may share the same failure path. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: diagnosis - single-path dependency screening. Decision interval: monthly and during resilience assessment. Minimum evidence: current verified dependency graph plus service criticality and health of each path. Method: identify services whose visibility/proof depends on one controller, gateway, network segment, sensor or meter and check for genuinely independent alternatives. Uncertainty: state documentation confidence and do not infer functional redundancy from duplicated point names. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Gives operators and engineers a practical shortlist of visibility blind spots to address in resilience planning.

Answer boundary

Do not redesign architecture, declare resilience compliance or disclose restricted network details beyond role permission. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

BMS/HVAC OPERATORS QUESTION

BM-080

What are this week's highest-priority verified HVAC energy and resilience issues, with evidence strength and action boundary?**Sensors and telemetry**

Signals / location: measured energy anomalies, unexplained runtime, degraded performance indicators, active resilience blind spots, current alarms/overrides and evidence-quality status. Coverage / timing: authorised weekly operational scope across verified HVAC assets and meter boundaries. native sources; aggregation: weekly issue register. Quality / provenance: do not combine incomparable metrics into a false precise score; formal safety and maintenance priorities always outrank optimisation analytics. Verify point or sensor type, physical/logical location and engineering units against the current mapping. Check source-clock alignment, last-good time, valid-sample uptime and gaps; for physical sensors/meters, check applicable calibration or field-check status and range/stuck/drift flags. Retain object ID, controller/gateway, mapping/configuration version, observed time and retrieved time; gaps stay missing.

Authoritative supporting data

Owners: the authorised BMS/energy-management owner, revenue or submeter owner, current resilience-strategy owner, weather-data owner, and asset/commissioning record owner. Precedence: emergency, life-safety, permit-to-work and resilience instructions > current approved source-of-record/control configuration > commissioned live telemetry > authorised notes > historical references. Preserve effective, observed and retrieved times, role permission, owner/version and provenance; show stale or conflicting sources separately and never reconcile them silently. The May 2021 floor plan is historical only and cannot establish current use, access, services, HVAC topology, hazards, closures or isolation boundaries.

Analysis required

Type: recommendation - weekly bounded prioritisation and evidence synthesis. Decision interval: weekly assessment. Minimum evidence: repeatable issue evidence, current approved operational constraints, source provenance and enough data to state practical impact without guessing. Method: group issues by energy, resilience and evidence quality; rank transparently by persistence, measured scope, service criticality and confidence. Uncertainty: show why each item is ranked, what remains unknown and which authorised role owns the next decision. Safe failure: if required mapping, permission, clock, health or sampling gates fail, return unknown / insufficient evidence and name the missing proof; do not interpolate missing control states or silently substitute a model.

Why this is useful

Creates an operator-ready weekly agenda focused on issues that are both important and actually evidenced.

Answer boundary

Do not promise savings/resilience outcome, authorise changes or replace formal risk, maintenance or emergency processes. No control writes, setpoint/schedule/priority changes, plant start/stop, override clearing/extending, alarm acknowledgement/suppression/reset, interlock bypass, work/access/isolation authorisation, or IAQ/comfort/compliance/safe-operation certification. Authorised operators, competent engineers and emergency/permit procedures decide and take precedence.

Catalogue summary

Eight complementary decision areas form one read-only operating catalogue. Use the narrowest question that matches the current decision and stop when minimum

1 Zone conditions, schedules and operating modes

Confirm scheduled mode, runtime windows, exceptional states and service relationships.

2 Setpoints, overrides and control intent

Trace effective intent, overrides, priorities, deadbands, adoption and loop saturation.

3 Temperature, humidity, air quality and comfort evidence

Assess measured environmental conditions without inferring people or certifying IAQ/comfort.

4 Plant, valves, dampers, fans and equipment status

Separate commands from run proof and screen actuator, flow, coil and cycling behaviour.

5 Alarms, faults, drift and sensor-quality diagnostics

Prioritise alarms and diagnose data-quality problems before trusting point values.

6 Control optimisation and bounded operational recommendations

Identify evidence-backed opportunities for authorised engineering assessment, never auto-control.

7 Maintenance, commissioning and shift handover

Prepare checks, call-outs, commissioning evidence and concise shift handovers.

8 Energy, peak demand, resilience and seasonal performance

Understand measured demand, runtime, seasonal performance and resilience visibility.

Operator use rules

Use current authorised sources first. Verify mappings, units, clocks and health before interpreting values. Preserve effective/observed/retrieved times and provenance. Keep gaps, staleness and conflicts visible. Separate direct observation from calculation, estimate, forecast, diagnosis and recommendation. Return unknown when minimum evidence fails. Keep all control, safety and certification decisions with authorised people and procedures.